Dr. Zhen Zhang

Associate Professor
Ph. D. Supervisor in Management Science

Data Science & Knowledge Management Group	Tel: +86-411 8470 9356
Institute of Systems Engineering	E-mail: zhen.zhang@dlut.edu.cn
School of Economics and Management	WWW: `https://desci-dut.github.io`
Dalian University of Technology, Dalian 116024, China	Last Updated: October 21, 2020

RESEARCH INTERESTS

- Group decision making, linguistic decision making and social network-based decision making
- Two-sided matching decision making, recommender systems
- Clustering, social network analysis

ACADEMIC WEBSITES

- **Publons**: https://publons.com/a/1408505
- **Google Scholar**: https://scholar.google.com/citations?user=vBSJplMAAAAJ
- **ORCID**: https://orcid.org/0000-0002-6512-1458
- **ResearchGate**: https://www.researchgate.net/profile/Zhen_Zhang64
- **Scopus ID**: https://www.scopus.com/authid/detail.uri?authorId=56900083800

EDUCATION

- **Visiting doctoral Student**, Centre for Quantum Computation & Intelligent Systems, University of Technology, Sydney, Australia, Oct. 2012 - Nov. 2013. **Supervisors: Prof. Jie Lu (IEEE/IFSA Fellow)& Guangquan Zhang**

- **Ph. D** in Management Science & Engineering, Institute of Systems Engineering, Dalian University of Technology, Dalian, China, Sep. 2008 - Oct. 2014. Supervisor: Prof. Chonghui Guo.
- **B.S. in Engineering Management**, School of Economics and Management, China University of Petroleum, Qingdao, China, Sept. 2004 - Jul. 2008, rank 1/61.

ACADEMIC EXPERIENCE

- **Deputy Director**, Institute of Systems Engineering, School of Economics and Management, Dalian University of Technology, Dalian, China, since Dec. 2019.
- **Associate Professor**, Institute of Systems Engineering, School of Economics and Management, Dalian University of Technology, Dalian, China, since Dec. 2018.
- **Core Member**, Center for Big Data and Intelligent Decision, Dalian University of Technology, Dalian, China, since Jan. 2019.
- **Core Member**, Dalian Key Laboratory of Data Science and Knowledge Management, Dalian University of Technology, Dalian, China, since Jan. 2017.
- **Visiting Scholar**, Department of Computer Science, University of Jaén, Spain, Nov. 2016 - Dec. 2016. **Supervisor: Luis Martínez**.
- **Postdoctoral Fellow in Business Administration**, Faculty of Management and Economics, Dalian University of Technology, Dalian, China, since Jan. 2015.
- **Assistant Professor**, Institute of Systems Engineering, Faculty of Management and Economics, Dalian University of Technology, Dalian, China, Nov. 2014 - Dec. 2018.

SELECTED HONORS AND AWARDS

- **Xinghai Outstanding Young Talents**, Dalian University of Technology, 2020.
- **Publons Global Peer Review Awards** - top 1% of reviewers in Computer Science and Cross-Field, 2019
- **Science and Technology Star for Young Scholars of Dalian City**, 2018
- **The Third Prize of Excellent Natural Science Academic Papers of Liaoning Province**, 2016, 2018, 2019.

- **The Third Prize of Excellent Natural Science Academic Papers of Dalian City**, 2016, 2017.
- **The First Prize of Excellent Natural Science Academic Papers of Dalian City**, 2015.
- **Outstanding Graduate of Liaoning Province**, 2014.
- **Outstanding Graduate of Dalian University of Technology**, 2013.
- **National Scholarship of China**, Dalian University of Technology, 2012.
- **Doctoral Student Academic Star** of Dalian University of Technology, 2012.
- **Noriko Mizuta Scholarship**, Dalian University of Technology, 2012.
- **Outstanding Doctoral Student** of Dalian University of Technology, 2011.

RESEARCH PROJECTS

[1] **PI**, Consensus-Oriented Models and Methods for Multi-Criteria Sorting Problems in Group Decision Making with Heterogeneous Information, granted by **National Natural Science Foundation of China** (71971039), CNY480K, 2020.1 - 2023.12.

[2] **PI**, Research on Two-sided Matching Decision Making Methods with Incomplete Information, granted by **National Natural Science Foundation of China** (71501023), CNY182.8K, 2016.1 - 2018.12.

[3] **PI**, Approaches to Two-sided Matching Decision Making with Incomplete Preference Information, granted by the **China Postdoctoral Science Foundation** (2015M570248), CNY80K, 2015.6 - 2017.6.

[4] **PI**, Consensus-Driven Models and Algorithms for Group Recommender Systems under Big Data environment, granted by *Dalian Science and Technology Bureau* (2018RQ69), CNY100K, 2019.1 - 2020.12

[5] **PI**, Consensus-Based Models for Linguistic Group Decision Making and Multi-Criteria Sorting, granted by **the Funds for Xinghai Outstanding Young Talents** (??????), CNY400K, 2020.1 - 2023.12.

[6] **PI**, Models and Methods for Linguistic Large-Scale Group Decision Making based on Social Network Trust, granted by **the Fundamental Research Funds for the Central Universities of China** (DUT17RC(4)11), CNY50K, 2017.7 - 2018.12.

[7] **PI**, Matching Decision Making Methods with Heterogeneous Criteria Sets for Matching Objects, granted by the **Fundamental Research Funds for the Central Universities of China** (DUT15RC(3)003), CNY50K, 2015.1 - 2016.12.

[8] **PI**, Group Decision Making with Uncertain Information, a Foundation for the Doctoral Student Academic Star of Dalian University of Technology, CNY10K, 2012.11 - 2013.10.

[9] **Participant**, Big-Data Driven Intelligent Decision Making Methods and Support Techniques for Smart City Construction in Dalian, a major program granted by Dalian Science and Technology Bureau (2018J11CY009), CNY3M, 2018.1 - 2020.12

[10] **Participant**, Clustering Models and Algorithms for Mining Electronic Medical Records, granted by **National Natural Science Foundation of China** (No. 71771034), CNY560K, 2018.1 - 2021.12.

[11] **Participant**, Evolutionary Clustering Models and Algorithms for Dynamic Data Mining, granted by **National Natural Science Foundation of China** (No. 71171030), CNY450K, 2012.1 - 2015.12.

[12] **Participant**, Information and Logistics Management in Emerging E-commerce, granted by the **Funds for Creative Research Groups of China** (No. 71421001), CNY8.4M, 2015.1 - 2020.12.

[13] **Participant**, Data Mining and Knowledge Discovery Based on Optimization and Multivariate Statistical Analysis: Methods and Applications, granted by the **Program for New Century Excellent Talents in University** (NECT-11-0050), CNY300K, 2012.1 - 2014.12.

[14] **Participant**, Clustering Models and Algorithms for Time Series Data Mining, granted by **National Natural Science Foundation of China** (No. 70871015), CNY230K, 2009.1 - 2011.12.

[15] **Participant**, Theory and Methods of Product Life Cycle-Oriented Knowledge Coordination Management, granted by the **Key program of National Natural Science Foundation of China** (No. 71031002), CNY1.5M, 2011.1 - 2014.12.

[16] **Participant**, Feature Representation and Similarity Measures for Time Series, granted by the **Fundamental Research Funds for the Central Universities of China** (DUT11SX04), CNY100K, 2011.1 - 2012.12.

[17] **Participant**, Path Selection and Policy Design for Regional Creativity Industry Development, granted by the **National Soft Science Research Program** (2013GXS2D018), CNY150K, 2014.1

- 2014.12.

[18] **Participant**, Enterprise Knowledge Management Methods and Enterprise Ontology, granted by the **National High Technology Research and Development Program** (2008AA04Z107), CNY710K, 2008.10 - 2010.12.

PUBLLICATIONS

Book Chapters [Total: 1]

[1] **Zhen Zhang**, Chonghui Guo. Association rules evaluation by a hybrid multiple criteria decision method, in: G. Yang (Ed.), *Multidisciplinary Studies in Knowledge and Systems Science*, IGI Global, Hershey PA, 2013, pp. 167-178. (Selected paper from the International Journal of Knowledge and Systems Science)

Journal Articles [Total: 26, SCI: 20, SSCI: 7, EI: 23]

[2] Yuzhu Wu, **Zhen Zhang**, Gang Kou, Hengjie Zhang, Xiangrui Chao, Cong-Cong Li, Yucheng Dong*, Francisco Herrera. Distributed linguistic representations in decision making: Taxonomy, applications, and challenges in data science and artificial intelligence, *Information Fusion (INFFUS)*, 2021, 65, pp.165-178, doi: 10.1016/j.inffus.2020.08.018. (SCI, EI, 2019 IF=13.669)

[3] **Zhen Zhang**, Xinyue Kou, Wenyu Yu*, Yuan Gao. Consistency improvement for fuzzy preference relations with self-confidence: An application in two-sided matching decision making, *Journal of the Operational Research Society (JORS)*, 2020, doi: 10.1080/01605682.2020.1748529. (SCI, SSCI, EI, 2019 IF=2.175, ABS 3*)

[4] **Zhen Zhang**, Wenyu Yu*, Luis Martínez, Yuan Gao. Managing multigranular unbalanced hesitant fuzzy linguistic information in multiattribute large-scale group decision making: A linguistic distribution-based approach, *IEEE Transactions on Fuzzy Systems*, 2019, doi: 10.1109/TFUZZ.2019.2949758. (SCI, EI, 2019 IF=9.518)

[5] Wenyu Yu, **Zhen Zhang***, Qiuyan Zhong. Consensus reaching for MAGDM with multi-granular hesitant fuzzy linguistic term sets: a minimum adjustment-based approach, *Annals of Operations Research*, 2019, doi: 10.1007/s10479-019-03432-7. (SCI, EI, 2019 IF=2.583, ABS 3*)

[6] **Zhen Zhang***, Yuan Gao, Zhuolin Li. Consensus reaching for social network group decision making by considering leadership and bounded confidence, *Knowledge-Based Systems (KBS)*,

2020, 204, 106240, doi: 10.1016/j.knosys.2020.106240. (SCI, EI, 2019 IF=5.921)

[7] **Zhen Zhang**, Xinyue Kou, Iván Palomares, Wenyu Yu*, Junliang Gao. Stable two-sided matching with incomplete fuzzy preference relations: A disappointment theory based approach, *Applied Soft Computing*, 2019, 84, 105730, doi: 10.1016/j.asoc.2019.105730. (SCI, SSCI, EI, 2018 IF=5.472)

[8] Yue Zheng, Guangquan Zhang, **Zhen Zhang**, Jie Lu*. A reducibility method for the weak linear bilevel programming problems and a case study in principal-agent, *Information Sciences (INS)*, 2018, 454-455, pp. 46-58, doi: 10.1016/j.ins.2018.04.045. (SCI, EI, 2018 IF=5.524)

[9] **Zhen Zhang***, Xinyue Kou, Wenyu Yu, Chonghui Guo. On priority weights and consistency for incomplete hesitant fuzzy preference relations, *Knowledge-Based Systems (KBS)*, 2018, 143, pp. 115-126, doi: 10.1016/j.knosys.2017.12.010. (SCI, EI, 2018 IF=5.101, **ESI hot paper**)

[10] **Zhen Zhang**, Xinyue Kou, Qingxing Dong*. Additive consistency analysis and improvement for hesitant fuzzy preference relations, *Expert Systems with Applications (ESWA)*, 2018, 98, pp. 118-128, doi: 10.1016/j.eswa.2018.01.016. (SCI, EI, 2018 IF=4.292, ABS 3*, **ESI highly cited paper**)

[11] Wenyu Yu, Qiuyan Zhong, **Zhen Zhang***. Multi-granular hesitant fuzzy linguistic group decision making with incomplete weight information, *Systems Engineering — Theory & Practice*, 2018, 38(3), pp. 777-785. (in Chinese, EI)

[12] **Zhen Zhang**, Chonghui Guo. Deriving priority weights from intuitionistic multiplicative preference relations under group decision-making settings, *Journal of the Operational Research Society (JORS)*, 2017, 68(12), pp. 1582-1599, doi: 10.1057/s41274-016-0171-6. (SCI, SSCI, EI, 2017 IF=1.396, ABS 3*)

[13] Wenyu Yu, **Zhen Zhang***, Qiuyan Zhong, Leilei Sun. Extended TODIM for multi-criteria group decision making based on unbalanced hesitant fuzzy linguistic term sets, *Computers & Industrial Engineering (CAIE)*, 2017, 114, pp. 316-328, doi: 10.1016/j.cie.2017.10.029. (SCI, EI, 2017 IF=3.195, ABS 2*, **ESI highly cited paper**)

[14] **Zhen Zhang**, Chonghui Guo, Luis Martínez. Managing multigranular linguistic distribution assessments in large-scale multiattribute group decision making, *IEEE Transactions on Systems, Man, and Cybernetics: Systems (TSMCS)*, 2017, 47(11), pp. 3063-3076, doi: 10.1109/TSMC.2016.2560521. (SCI, SSCI, EI, **2017 IF=5.131, ESI highly cited paper**)

[15] **Zhen Zhang**, Chonghui Guo. A VIKOR-based approach to group decision making with uncertain preference ordinals and incomplete weight information, *Informatica*, 2016, 27(3), pp. 689-708, doi: 10.15388/Informatica.2016.106. (SCI, SSCI, EI, 2016 IF=1.056)

[16] **Zhen Zhang**, Chonghui Guo. Fusion of heterogeneous incomplete hesitant preference relations in group decision making, *International Journal of Computational Intelligence Systems (IJCIS)*, 2016, 9(2), pp. 245-262, doi: 10.1080/18756891.2016.1149999. (SCI, EI, 2016 IF=1.14)

[17] **Zhen Zhang**, Chonghui Guo. Notes on "logarithmic least squares method to priority for group decision making with incomplete fuzzy preference relations", *Applied Mathematical Modelling (AMM)*, 2016, 40(3), pp. 1788-1792, doi: 10.1016/j.apm.2015.07.024. (SCI, EI, 2016 IF=2.35)

[18] **Zhen Zhang**, Chonghui Guo. Consistency and consensus models for group decision-making with uncertain 2-tuple linguistic preference relations, *International Journal of Systems Science (IJSS)*, 2016, 47(11), pp. 2572-2587, doi: 10.1080/00207721.2014.999732. (SCI, SSCI, EI, 2016 IF=2.285)

[19] **Zhen Zhang**, Chonghui Guo. Consistency-based algorithms to estimate missing elements for uncertain 2-tuple linguistic preference relations, *International Journal of Computational Intelligence Systems (IJCIS)*, 2014, 7(5), pp. 924-936, doi: 10.1080/18756891.2013.856254. (SCI, EI, 2014 IF=0.574)

[20] **Zhen Zhang**, Chonghui Guo. An approach to group decision making with heterogeneous incomplete uncertain preference relations, *Computers & Industrial Engineering (CAIE)*, 2014, 71, pp. 27-36, doi: 10.1016/j.cie.2014.02.004. (SCI, EI, 2014 IF=1.783)

[21] Chonghui Guo, Jiajia Wang, **Zhen Zhang**. Evolutionary community structure discovery in dynamic weighted networks, *Physica A*, 2014, 413, pp. 565-576, doi: 10.1016/j.physa.2014.07.004. (SCI, EI, 2014 IF=1.732)

[22] **Zhen Zhang**, Chonghui Guo. A method for multi-granularity uncertain linguistic group decision making with incomplete weight information, *Knowledge-Based Systems (KBS)*, 2012, 26, pp. 111-119, doi: 10.1016/j.knosys.2011.07.009. (SCI, EI, **2012 IF=4.104**)

[23] **Zhen Zhang**, Chonghui Guo. A multi-granularity uncertain linguistic group decision making method based on relative entropy, *Journal of Dalian University of Technology*, 2012, 52(6), pp. 921-927. (in Chinese, EI)

[24] **Zhen Zhang**, Chonghui Guo. Association rules evaluation by a hybrid multiple criteria decision method, *International Journal of Knowledge and Systems Science (IJKSS)*, 2011, 2(3), pp. 14-25. (INSPEC)

[25] **Zhen Zhang**, Chonghui Guo. Multiple attributes group decision making method based on two-tuple linguistic information processing, *Control and Decision*, 2011, 26(12), pp. 1881-1885. (in Chinese, EI)

[26] Chonghui Guo, **Zhen Zhang**. Interestingness evaluation of association rules based on combination evaluation method, *Journal of the China Society for Scientific and Technical Information*, 2011, 30(4), pp. 353-360. (in Chinese, CSSCI)

[27] Shanglei Chai, Chonghui Guo, **Zhen Zhang**. Asymmetric and heteroskedastic model of volatility of international stock index and its clustering analysis, *Journal of Systems and Management*, 2011, 20(2), pp. 136-142. (in Chinese)

Conference Papers [Total: 15, EI: 14]

[28] **Zhen Zhang**, Zhuolin Li, Wenyu Yu*. Some models to improve the worst consistency level of hesitant fuzzy preference relations. In Proceedings of *the 14th International FLINS Conference on Robotics and Artificial Intelligence (FLINS 2020)*, pp. 327-335, Cologne, Germany, 18-21 August 2020. (EI, ISTP)

[29] Amin Mahmoudi, Saad Ahmed Javed, **Zhen Zhang***, Xiaopeng Deng. Grey group QUALIFLEX method: Application in project management. In Proceedings of *the 14th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2019)*, pp. 189-195, Dalian, China, 14-16 November, 2019. (EI, ISTP)

[30] **Zhen Zhang**, Yuan Gao, Wenyu Yu. Group decision making with linguistic distribution preference relations: An interval numerical scale-based approach. In Proceedings of *the 14th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2019)*, pp. 125-128, Dalian, China, 14-16 November, 2019. (EI, ISTP)

[31] **Zhen Zhang**, Junliang Gao, Wenyu Yu. Evaluating collaboration ability of R&D team members: A multi-granular hesitant fuzzy linguistic approach. In Proceedings of *the 11th Conference of the European Society for Fuzzy Logic and Technology (EUSFLAT 2019)*, pp. 30-36, Prague, 9-13 September, 2019. (EI, ISTP)

[32] **Zhen Zhang**, Wenyu Yu*. An optimization-based approach to aggregating multi-granular hesitant fuzzy linguistic term sets. In Proceedings of *the 13th International FLINS Conference on Data Science and Knowledge Engineering for Sensing Decision Support (FLINS 2018)*, pp. 205-212, Belfast, Northern Ireland, UK, 21-24, August 2018. (ISTP)

[33] **Zhen Zhang**, Xinyue Kou, Wenyu Yu. Two-sided matching decision making based on heterogeneous incomplete preference relations. In Proceedings of *the 12th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2017)*, pp. 1-6, Nanjing, China, 24-26 November, 2017. (EI)

[34] Wei Wei, Chonghui Guo, Jingfeng Chen, **Zhen Zhang**. Textual topic evolution analysis based on term co-occurrence: A case study on the government work report of the State Council (1954-2017). In Proceedings of *the 12th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2017)*, pp. 1-6, Nanjing, China, 24-26 November, 2017. (EI)

[35] Wenyu Yu, **Zhen Zhang***, Qiuyan Zhong. A TODIM-based approach to large-scale group decision making with multi-granular unbalanced linguistic information. In *Proceedings of the 2017 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, pp. 1-6, Naples, Italy, 9-12 July, 2017. (one of the 11 best papers selected from 198 student papers submitted to Fuzz-IEEE 2017, ERA tier A conference, EI)

[36] **Zhen Zhang**, Wenyu Yu, Chonghui Guo. Group decision making based on acceptably consistent interval multiplicative preference relations. In Proceedings of the 17th International Symposium on Knowledge and Systems Sciences (KSS 2016), published by CCIS (vol. 660), pp. 165-174, Kobe, Japan, 4-6 November, 2016. (EI)

[37] **Zhen Zhang**, Chonghui Guo. Minimum adjustment-based consistency and consensus models for group decision making with interval pairwise comparison matrices. In *Proceedings of the 2016 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, pp. 1701-1708, Vancouver, Canada, 24-29 July, 2016. (ERA tier A conference, EI)

[38] Wenyu Yu, Qiuyan Zhong, **Zhen Zhang***. Fusing multi-granular unbalanced hesitant fuzzy linguistic information in group decision making. In *Proceedings of the 2016 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, pp. 872-879, Vancouver, Canada, 24-29 July, 2016. (ERA tier A conference, EI)

[39] **Zhen Zhang**, Chonghui Guo. New operations of hesitant fuzzy linguistic term sets with applications in multi-attribute group decision making. In *Proceedings of the 2015 IEEE International*

Conference on Fuzzy Systems (FUZZ-IEEE 2015), pp. 1-8, Istanbul, Turkey, 2-5 August, 2015. (ERA tier A conference, EI)

[40] **Zhen Zhang**, Guangquan Zhang, Jie Lu, Chonghui Guo. A fuzzy tri-level decision making algorithm and its application in supply chain. In *Proceedings of the 8th conference of the European Society for Fuzzy Logic and Technology (EUSFLAT 2013)*, pp. 154-160, Milan, Italy, 11-13 September, 2013. (EI)

[41] **Zhen Zhang**, Chonghui Guo. A hybrid multiple attributes two-sided matching decision making method with incomplete weight information. In *Proceedings of the 11th International Workshop on Meta-synthesis and Complex Systems (MCS 2011)*, published by LNAI (vol. 6889), pp. 272-283, Lanzhou, China, 7-9 September, 2011. (EI)

[42] **Zhen Zhang**, Chonghui Guo. Ranking association rules by TOPSIS method with combination weights. In *Proceedings of the 10th International Symposium on Knowledge and Systems Sciences (KSS 2009)*, Hong Kong, China, 3-4 December 2009. (CD-ROM).

Technical Reports [Total: 3]

[43] **Zhen Zhang**, Guangquan Zhang, Jie Lu, Chonghui Guo. An extended Kth-best algorithm for linear tri-level programming with multi-follower uncooperative relationships, technical report, Aug. 2013.

[44] **Zhen Zhang**, Chonghui Guo. Hesitant multiplicative preference relation and its application in group decision making, technical report, Nov. 2013.

[45] **Zhen Zhang**, Chonghui Guo. A note on "An automatic approach to reaching consensus in multiple attribute group decision making", technical report, Jul. 2011.

Courses Teaching

- **Complex Network: Theories and Methods**, Postgraduates, Autumn 2018, 2019, 2020
- **Business Statistics and R**, Undergraduates, Autumn 2017, 2018, 2019, 2020
- **Principles of Management**, Undergraduates, Autumn 2017, 2018, 2019, 2020
- **Information and Knowledge Management**, Postgraduates, Spring 2016
- **Introduction to Knowledge Management**, Undergraduates, Summer 2016

Supervised Students

Master Students

- Nan Yuan and Xiaoyun Zhang, to be enrolled in Sep. 2021
- Jinfang Gu and Shuyao Shen, enrolled in Sep. 2020
- Zhuolin Li, Siyuan Ji and Nan Wang, enrolled in Sep. 2019
- Yuan Gao, enrolled in Sep. 2018
- Junliang Gao, graduated in June 2020
- Xinyue Kou, graduated in June 2019

MBA/MEM/MPM Students

- Chunyang Huang, graduated in 2020
- Peng Jiang and Xiangli Liu, graduated in 2019
- Wei Liu, Jiahe She, Lei Wang and Jiahui Zhang, graduated in 2018
- Shuo Li, graduated in 2017
- Wen Gu, Naizhe Liu, Yong Peng, Hengyue Sun, Yaowei Wang, Congqin Zhang, Shuang Zhang and Weijing Zhu, enrolled in 2019
- Lu Bai, Ji He, Qing Li, Tiejian Wang, Xiamei Xu and Shaoqun Yang, enrolled in 2018
- Yumeng Li, Lu Wang and Qingyuan Yang, enrolled in 2017

Undergraduate Students

- Zihan Xia, enrolled in Sep. 2019
- Jialu Wang, enrolled in Sep. 2018
- Mian Wu, enrolled in Sep. 2017
- Miao Chen, Shuai Liu, Yujian Qiu, Ruiying Yang and Zhenyu Yao (thesis supervision in 2020, graduated in 2020)
- Lin Liu and Xiankun Peng (thesis supervision in 2020), graduated in 2020
- Wenwen Ma and Yuexin Wang, graduated in 2019
- Sichen Tan and Jiayi Yang, graduated in 2018

- Jian Cheng, Jiaxu Liu, Zhemin Liu, Bingshuo Ren and Xinyi Yu (thesis supervision in 2019, graduated in 2019)
- Xinze Chang, Danyang Song, Jia Song and Shuting Zhang (thesis supervision in 2018, graduated in 2018)
- Min Chen, Junpeng Sun, Jinxiao Tan, Yao Yu and Kunbo Zhang (thesis supervision in 2017, graduated in 2017)

PROFESSIONAL ACTIVITIES

Consultancies and Developments

- **Data Processing and Analytics for Dalian Land Resources Bureau**, Co-PI, Sep. 2017 - Oct. 2017.
  Responsibilities: a. Data preprocessing and visualization; b. Extracting key indexes for house price prediction; c. Building models for house price prediction.
- **Test Bank Resources Data Analysis Based on Big Data Analytics Techniques**, Co-PI, May 2017 - Oct. 2018.
  Responsibilities: a. Knowledge points tagging for test bank resources data; b. Test question personalized recommendation based on test bank resources data.
- **Big Data-Driven Key Techniques for Payment Channels Data Analytics in Electric Power Industry**, Co-PI, Apr. 2016 - Mar. 2017.
  Responsibilities: a. Building models for payment channels design; b. Building GIS-based site selection models; c. Building models for user credit rating.
- **Engine Parameters Data Mining**, Jan. 2011 - Dec. 2011.
  Responsibilities: a. Building prediction models for engine parameters with BP neural network; b. System Development; c. Composing technical report for the whole project.
- **Customer Experience Management-Oriented Data Mining for Automobile 4S Shop**, Oct. 2010 - Jan. 2011.
  Responsibilities: a. Building customer churn prediction models; b. Predicting customer churn status with real-world customer data; c. Composing technical report for the whole project.

Memberships & Professions

- Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, IEEE Computational Intelligence Society
- Member, IEEE Systems, Man, and Cybernetics Society
- Member, European Society for Fuzzy Logic and Technology (EUSFLAT)
- Member, Systems Engineering Society of China (SESC)
- Member, Operations Research Society of China (ORSC)
- Member, China Computer Federation (CCF)
- Member, Chinese Society of Optimization, Overall Planning and Economic Mathematics (CSOOPEM)
- Committee member and Vice secretary-general, Data Science and Knowledge Systems Engineering Branch, SESC
- Board member, Intelligent Decision Making and Game Branch, CSOOPEM
- Committee member, the 3rd Youth Working Committee, CSOOPEM
- Member, Chinese Association of Automationof China (CAA)
- Vice secretary-general, Business Intelligence Professional Committee of the Chinese Society for Management Modernization (CSMM) (Since 2016)
- Reviewer for the National Nature Science Foundation of China (NSFC) (Since 2018)

Editorial Boards

- **Associate Editor** of Journal of Intelligent & Fuzzy Systems (SCI, IF: 1.851), since Dec. 2018
  https://www.iospress.nl/journal/journal-of-intelligent-fuzzy-systems/?tab=editorial-board
- **Associate Editor** of Kybernetes (SCI, IF: 1.754), since Jan. 2019
  http://www.emeraldgrouppublishing.com/products/journals/editorial_team.htm?id=k
- **Editorial Board Member** of International Journal of Computational Intelligence Systems (SCI, IF: 1.838), since Nov. 2019
  https://www.atlantis-press.com/journals/ijcis/editorial-board
- **Associate Editor** of Frontiers in Artificial Intelligence (Fuzzy Systems Section), since Jan. 2020
  https://www.frontiersin.org/journals/artificial-intelligence/sections/fuzzy-systems#editorial-board

- **Associate Editor** of Frontiers in Applied Mathematics and Statistics (Optimization Section), since May 2020
  https://www.frontiersin.org/journals/applied-mathematics-and-statistics/sections/optimization#editorial-board
- **Review Editor** of Frontiers in Big Data, since Jan. 2019
  https://www.frontiersin.org/journals/big-data#editorial-board
- **Editor** of ICSES Transactions on Neural and Fuzzy Computing, since April 2019
  http://www.i-cses.com/itnfc/index.php#anchor1
- Secretary of the Big Data and Knowledge Management Branch Editorial Board, Management Science and Engineering Volume, Encyclopedia of China (The 3rd edition)

Reviews & Conference Service

- **Journal Reviewers for**
  - Advances in Fuzzy Systems (SCI)
  - African Journal of Business Management (SSCI)
  - Algorithms (SCI)
  - Annals of Operations Research (SCI)
  - Applied Intelligence (SCI)
  - Applied Soft Computing (SCI)
  - Artificial Intelligence in Medicine (SCI)
  - Artificial Intelligence Review (SCI)
  - Asia-Pacific Journal of Operational Research (SCI)
  - Axioms (ESCI)
  - British Journal of Mathematical and Statistical Psychology (SCI/SSCI)
  - Computers & Industrial Engineering (SCI)
  - Decision Sciences (SSCI)
  - Electronics (SCI)
  - European Journal of Operational Research (SCI)
  - Expert Systems (SCI)

- Fuzzy Optimization and Decision Making (SCI)
- Fuzzy Sets and Systems (SCI)
- Group Decision and Negotiation (SSCI)
- Hacettepe Journal of Mathematics and Statistics (SCI)
- Healthcare (SCI)
- IEEE Access (SCI)
- IEEE Transactions on Cybernetics (SCI)
- IEEE Transactions on Engineering Management (SCI/SSCI)
- IEEE Transactions on Fuzzy Systems (SCI)
- IEEE Transactions on Systems, Man, and Cybernetics: Systems (SCI)
- IEEE Transactions on Reliability (SCI)
- IET Information Security (SCI)
- INFORMATICA (SCI)
- Information-MDPI (ESCI, EI)
- Information Fusion (SCI)
- Information Sciences (SCI)
- Information Systems Frontiers (SCI)
- International Journal of Computational Intelligence Systems (SCI)
- International Journal of Disaster Risk Reduction (SCI)
- International Journal of Environmental Research and Public Health (SCI)
- International Journal of Fuzzy System Applications
- International Journal of Fuzzy Systems (SCI)
- International Journal of Hospitality Management (SSCI)
- International Journal of Information Technology & Decision Making (SCI)
- International Journal of Intelligent Computing and Cybernetics (ESCI)
- International Journal of Intelligent Systemsă(SCI)
- International Journal of Machine Learning and Cybernetics (SCI)

- International Journal of Production Research (SCI)
- International Journal of Systems Science: Operations & Logistics
- International Journal of Systems Science (SCI)
- International Journal of Uncertainty, Fuzziness and Knowledge-Based System (SCI)
- Iranian Journal of Fuzzy Systems (SCI)
- Journal of Ambient Intelligence and Humanized Computing (SCI)
- Journal of Applied Statistics (SCI)
- Journal of Experimental & Theoretical Artificial Intelligence (SCI)
- Journal of Cleaner Production (SCI)
- Journal of Intelligent & Fuzzy Systems (SCI)
- Journal of Systems Science and Complexity (SCI)
- Journal of the Operational Research Society (SCI/SSCI)
- Knowledge-Based Systems (SCI)
- Kybernetes (SCI)
- Mathematical Problems in Engineering (SCI)
- Natural Hazards (SCI)
- Neurocomputingă(SCI)
- Omega (SCI/SSCI)
- Physica A (SCI)
- PLOS ONEă(SCI)
- Soft Computing (SCI)
- Symmetry (SCI)
- Technological and Economic Development of Economy (SSCI)
- The Scientific World Journal (SCI)
- Journal of Computational Design and Engineering (Scopus)
- OPSEARCH (Scopus)
- The Open Cybernetics & Systemics Journal (Scopus)

- Annals of Fuzzy Mathematics and Informatics
- Granular Computing
- Heliyon
- J-MDPI
- Journal of Systems Science and Information
- Control and Decision (EI)
- Systems Engineering - Theory & Practice (EI)
- Systems Engineering and Electronics (EI)
- Journal of Industrial Engineering and Engineering Management
- Journal of Systems Engineering
- Nankai Business Review
- Journal of University of Science and Technology of China
- Journal of Systems Science and Mathematical Sciences
- Journal of Mathematics in Practice and Theory
- Journal of Safety Science and Technology

- **PC Members & Reviewers for**
  - The 2020 International Joint Conference on Neural Networks (IJCNN 2020)
  - The 2020 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2020)
  - The 14th International FLINS Conference on Robotics and Artificial Intelligence (FLINS 2020)
  - The 20th International Conference on Group Decision and Negotiation (GDN 2020)
  - The 17th International Conference on Distributed Computing and Artificial Intelligence (DCAI 2020)
  - The 20th International Symposium on Knowledge and Systems Sciences (KSS 2019)
  - The 14th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2019)
  - The 1st International Alan Turing Conference on Decision Support and Recommender Systems (DSRS-Turing 2019)

- The 2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2019)
- Joint 10th International Conference on Soft Computing and Intelligent Systems and 19th International Symposium on Advanced Intelligent Systems in conjunction with Intelligent Systems Workshop 2018 (SCIS-ISIS2018)
- The 2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC 2018)
- The 2018 IEEE International Conference on Technology Management, Operations and Decisions (IEEE ICTMOD 2018)
- The 19th International Symposium on Knowledge and Systems Sciences (KSS 2018)
- The 37th North American Fuzzy Information Processing Society Annual Conference (NAFIPS 2018)
- The 13th International FLINS Conference on Data Science and Knowledge Engineering for Sensing Decision Support (FLINS 2018)
- The 2018 IEEE World Congress on Computational Intelligence (WCCI 2018)
- The 18th International Conference on Group Decision and Negotiation (GDN 2018)
- The 18th International Symposium on Knowledge and Systems Sciences (KSS 2017)
- The 12th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2017)
- Joint 17th World Congress of International Fuzzy Systems Association and 9th International Conference on Soft Computing and Intelligent Systems (IFSA-SCIS 2017)
- The 2017 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2017)
- The 17th International Symposium on Knowledge and Systems Sciences (KSS 2016)
- The 12th International FLINS Conference on Decision Making and Soft Computing (FLINS 2016)
- The 2016 IEEE World Congress on Computational Intelligence (WCCI 2016)
- The 16th World Congress of the International Fuzzy Systems Association and the 9th Conference of the European Society for Fuzzy Logic and Technology (IFSA-EUSFLAT 2015)
- The 16th International Symposium on Knowledge and Systems Sciences (KSS 2015)
- The 11th International FLINS Conference on Decision Making and Soft Computing (FLINS 2014)

- The 2013 International Conference on Knowledge Science, Engineering and Management (KSEM 2013)
- The 15th IFSA World Congress and 32nd Annual Meeting of NAFIPS (IFSA-NAFIPS 2013)
- The 8th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2013)
- The 12th International Workshop on Meta-synthesis & Complex Systems (MCS 2012)

- **Special Session/Stream Organizers for**
  - "Consensus for large-scale Group Decision Making" in the 2020 International Conference on System Science and Engineering 2020 (ICSSE 2020), co-organized by Jian Wu, Yucheng Dong, Zaiwu Gong, Yejun Xu and Zhen Zhang.
  - "Managing Uncertaity in Data-Driven Decision Making" in the 14th International FLINS Conference on Robotics and Artificial Intelligence (FLINS 2020), co-organized by Zhen Zhang, Yucheng Dong and Yejun Xu.
  - "Intelligent Group Decision Making and Consensus Process" in the 20th International Conference on Group Decision and Negotiation (GDN 2020), co-organized by Zhen Zhang, Yucheng Dong, Francisco Chiclana and Enrique Herrera-Videma.
  - "Data-Driven Decision Making with Uncertainty" in the 14th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2019), co-organized by Luis Martínez, Yucheng Dong and Zhen Zhang.
  - "Group Decision Making in Social Networks" in the 49th International Conference on Computers & Industrial Engineering (CIE 49), co-organized by Jian Wu, Yucheng Dong, Zaiwu Gong, Yejun Xu and Zhen Zhang.
  - "Consensus Process Decision Making" in the 18th International Conference on Group Decision and Negotiation (GDN 2018), co-organized by Yucheng Dong, Zaiwu Gong, Jian Wu, Zhen Zhang and Haiming Liang.
  - "Intelligent Decision Support Systems: Models, Methods and Applications" in the 12th International Conference on Intelligent Systems and Knowledge Engineering (ISKE 2017), co-organized by Zhen Zhang and Jialin Han.